Lecture 11

18.11.2025

For convergence, we need the following estimates.

Theorem (estimates) Let $M \in \mathbb{R}^{d\times d}$ be s.t. $M_{ii} > 0$, $M_{ij} \le 0$, $i \neq j$ and $\sum_{j=1}^{d} M_{ij} \ge 0$ for all $i = 1, \dots, d$ ("weakly diagonally dominant"). Then

i) $\|v\|_\infty \le \|(I_d + M)v\|_\infty$.

ii) If $0 < M_{ii} \le 1$, then $\|(I_d - M)v\|_\infty \le \|v\|_\infty$.

Proof

i) Let $i \in \{1, \dots, d\}$ be s.t. $\|v\|_\infty = |v_i| \neq 0$.

$$\Rightarrow \quad \|(I_d + M)v\|_\infty \ge |((I_d+M)v)_i| = \left| \sum_{j=1}^{d} (I_d + M)_{ij} v_j \right|$$

$$= \left| 1 + \sum_{j=1}^{d} M_{ij} \frac{v_j}{v_i} \right| \cdot |v_i| = \left| 1 + M_{ii} + \sum_{\substack{j=1 \\ j\neq i}}^{d} \underbrace{M_{ij}}_{\le 0} \underbrace{\frac{v_j}{v_i}}_{\in[-1,1]} \right| \cdot |v_i|$$

$$\ge \Big| 1 + \underbrace{M_{ii} + \sum_{j\neq i} M_{ij}}_{\ge 0} \Big| \cdot |v_i| \ge |v_i| = \|v\|_\infty .$$

ii) $$\|(I_d - M)v\|_\infty = \max_i \left| v_i - M_{ii} v_i - \sum_{j\neq i} M_{ij} v_j \right|$$

$$= \max_i \left| \underbrace{(1 - M_{ii})}_{\ge 0} v_i + \sum_{j\neq i} \underbrace{(-M_{ij})}_{\ge 0} v_j \right|$$

$$\le \max_i \; (1 - M_{ii}) \|v\|_\infty + \sum_{j\neq i} (-M_{ij}) \|v\|_\infty = \max_i \|v\|_\infty \underbrace{- \sum_{j=1}^{d} M_{ij} \|v\|_\infty}_{\le 0} \le \|v\|_\infty$$

$\blacksquare$

Recall stability for $L_h u_h = f_h$, we called the scheme stable if $|||L_h^{-1}||| < c$ for some constant c which is independent of h

In other words: $\|u_h\|_h = \|L_h^{-1} f_h\|_h \le c \|f_h\|_h$ so the discrete solution u_h should be bounded by a constant (indp. of h!) and the norm of the given data f_h

$\rightsquigarrow$ we will use this interpretation of stability and show that the discrete solution of the θ-scheme is bounded in terms of the given data, namely the initial condition v_0 and the right-hand side f

We can rewrite the θ-scheme (in vector form) as

$$u_h^{k+1} + \tau\,\theta\, L_h\, u_h^{k+1} = \tau\, f_h^k + (I - \tau(1-\theta) L_h)\, u_h^k \qquad (*)$$

We first apply estimate i) to the matrix $M = \tau\,\theta\, L_h$ (assumptions true!) and obtain

$$\|u_h^{k+1}\|_\infty \overset{i)}{\leq} \|(I+M)\,u_h^k\|_\infty \overset{(*)}{=} \|\tau\, f_h^k + (I - \tau(1-\theta) L_h)\, u_h^k\|_\infty$$

$$\leq \tau\, \|f_h^k\|_\infty + \|(I - \underbrace{\tau(1-\theta) L_h}_{=\tilde{M}})\, u_h^k\|_\infty$$

IF $0 < \tilde{M}_{ii} \leq 1$, we can also apply estimate ii) of the theorem to obtain

$$\|u_h^{k+1}\|_\infty \leq \tau\, \|f_h^k\|_\infty + \|(I - \tilde{M})\, u_h^k\|_\infty \overset{ii)}{\leq} \tau\, \|f_h^k\|_\infty + \|u_h^k\|_\infty$$

$$\underset{\text{as before}}{\overset{\text{again}}{\leq \dots \leq}} \underbrace{\|u_h^0\|_\infty}_{\text{initial data}} + \tau \sum_{j=0}^{k} \underbrace{\|f_h^j\|_\infty}_{\text{right hand side}} \quad \Rightarrow \text{ stability!}$$

When do we have $0 < \tilde{M}_{ii} \leq 1$? $\quad 0 < \tau(1-\theta)\frac{2}{h^2} \leq 1 \Rightarrow (1-\theta)\tau \leq \frac{h^2}{2}$

always true for $\theta = 1$!

Altogether we now have shown (consistency + stability $\Rightarrow$ convergence) the following result:

<u>Theorem</u> Let $(1-\theta)\tau \leq \frac{h^2}{2}$ and $u \in C^{\overset{\text{time}}{2},\overset{\text{space}}{4}}$. Then

$$\max_k \|u_h^k - R_h u^k\|_\infty = \mathcal{O}(\tau + h^2).$$

If $\theta = \frac{1}{2}$ with $\tau \leq h^2$ and $u \in C^{3,4}$ then

$$\max_k \|u_h^k - R_h u^k\|_\infty = \mathcal{O}(\tau^2 + h^2).$$

The finite element method

Recall we know that the finite difference method converges *if* the exact solution u has a certain regularity, e.g., $u \in C^{2,4}$ for the heat equation

BUT what if we do not have this regularity?

Motivating example consider an equation for the electric potential u given by

$$-(a(x)\,u'(x))' = f(x) \qquad \text{for } x \in (0,2)$$

$$u(0) = u_0, \quad u(2) = u_2$$

where f is a charge distribution and the dielectric constant a is discontinuous, e.g.

$$a(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 2 & 1 < x \le 2 \end{cases}$$

$\rightsquigarrow$ for example if material in $(0,1)$ has different properties than the one in $(1,2)$

Claim the homogeneous problem ($f \equiv 0$) has the solution

$$u(x) = u_0 + \begin{cases} \alpha x & 0 \le x \le 1 \\ \alpha + \beta(x-1) & 1 < x \le 2 \end{cases}$$

with $\alpha + \beta = u_2 - u_0$ and $\alpha - 2\beta = 0$ characterizing α, β.

Proof (claim)

$$u'(x) = \begin{cases} \alpha & 0 \le x \le 1 \\ \beta & 1 < x \le 2 \end{cases}$$

Note u' is not continuous if $\alpha \neq \beta$!

$$\Rightarrow \quad a(x)\,u'(x) = \begin{cases} \alpha & 0 \le x \le 1 \\ \overset{\shortparallel}{2\beta} & 1 < x \le 2 \end{cases} \quad \Rightarrow \quad -(a(x)u'(x))' = 0 \checkmark$$

boundary conditions satisfied? $u(0) = u_0 \checkmark$ $\quad u(2) = u_0 + \alpha + \beta(2-1)$

$= u_2 \checkmark$

Why do we need $\alpha - 2\beta = 0$? $\rightsquigarrow$ $a(x)\,u'(x)$ becomes differentiable!

Conclusion although $u \notin C^1(0,2)$, we can still define u such that the equation is satisfied pointwise!

$\rightsquigarrow$ maybe our initial concept of a solution $u \in C^2(\Omega) \cap C(\bar{\Omega})$ was too strict?

$\rightarrow$ do we have weaker concepts?

Weak solutions and variational problems

Let us again consider $\quad -u''(x) = f(x) \quad$ in $\Omega = (0,1) \qquad (*)$

$$u(0) = 0 = u(1)$$

Given an arbitrary continuously differentiable function $v \in C^1(\bar{\Omega})$, let us multiply $(*)$ by v and integrate over Ω:

$$\Rightarrow \qquad -\int_\Omega u''(x)v(x)\,dx = \int_\Omega f(x)v(x)\,dx$$

With integration by parts, we equivalently obtain

$$\int_\Omega f(x)v(x)\,dx = -\int_\Omega u''(x)v(x)\,dx \overset{\text{i.b.p.}}{=} \int_\Omega u'(x)v'(x)\,dx - \left[u'(x)v(x)\right]_0^1$$

Since $v \in C^1(\bar{\Omega})$ was arbitrary, this equality holds for all $v \in C^1(\bar{\Omega})$!

In particular, we can do the same for $v \in C_0^1(\bar{\Omega}) := \{w \in C^1(\bar{\Omega}) \mid w(0) = 0 = w(1)\}$.

$$\Rightarrow \int_\Omega f(x)v(x)\,dx = \int_\Omega u'(x)v'(x)\,dx - \underbrace{\left[u'(x)\,v(x)\right]_0^1}_{\text{vanishes!}} = \int_\Omega u'(x)v'(x)\,dx$$

Instead of looking for $u \in C^2(\Omega) \cap C(\bar{\Omega})$ satisfying $(*)$, let us consider:

$$\boxed{\text{Find } u \in C_0^1(\bar{\Omega}) \text{ s.t. } \int_\Omega u'(x)v'(x)\,dx = \int_\Omega f(x)v(x)\,dx \text{ for all } v \in C_0^1(\bar{\Omega})}$$

We will refer to this problem (formulation) as variational problem or weak problem